Telling Categories of Unexploded Ordnance Apart in Photos: A CNN Trained With and Without Hidden Patches

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Report Overview

In this project I trained a convolutional neural network (CNN) in PyTorch to sort photos of unexploded ordnance (UXO) into five categories, projectile, grenade, mortar bomb, cartridge and aviation bomb, as a first indication that an explosive ordnance disposal (EOD) technician would then confirm. Such a tool would generally be used only in war-torn areas or on former battlefields. I used CTX-UXO, a public dataset of 3,520 photos of real scenes in which 15,449 items are labelled (Craioveanu et al., 2026), and classified 64 × 64-pixel crops cut around the labelled items. I compared a small CNN trained from scratch with an otherwise identical model trained with random erasing, which hides a random part of each training image (Zhong et al., 2020), because in the field the features that tell items apart are often hidden by brush or partial burial.

Dataset and Task Description

The Dataset

CTX-UXO was collected from October 2021 to July 2025 by researchers of the National University of Science and Technology Politehnica Bucharest with the help of pyrotechnical personnel of the Romanian Inspectorate for Emergency Situations, in all seasons and kinds of weather, and most of its photos were taken with a phone camera (Craioveanu et al., 2026). The items come from the historical conflicts of Europe, and the photos show them in real settings, in grass, leaves, soil and snow, sometimes gathered in rows and piles (Figure 1); inert items and replicas were also photographed from several angles during training sessions for EOD operators. One trained UXO expert labelled each item with a box and one of 12 categories. I used version 1

of the dataset on Zenodo, published under a CC BY 4.0 licence and downloaded on October 20, 2026 (Craioveanu & Stamatescu, 2025); the repository's README gives its address and checksum, and the notebook can rebuild the crops from it.

Figure 1

Six Training Photos With Their Labelled Boxes



Note. From notebook Section 2.2. The photos are shrunk for display. Photos from CTX-UXO (Craioveanu & Stamatescu, 2025), CC BY 4.0.

The creators split the photos into training (2,461), validation (530) and test (529) folders, so all the items in one photo stay in one split. Five categories hold 15,109 of the 15,449 labels (98%), and I kept those; the other seven, with 5 to 124 labels each, are too few to learn or test, and I set their 340 crops aside to see what the models say about kinds of items they never saw. The five are unequal, with 18 projectiles for every aviation bomb (Table 1). Before relying on the labels, I checked two category names: the dataset record gives 4,269 mortar bombs and 3,399

grenades and the label files the opposite, and a blind check, in which I said which of two unlabelled sets of 12 crops showed grenades and which mortar bombs, confirmed the label files.

Table 1

Crops of the Five Categories Kept, by Split

Category	Training crops	Validation crops	Test crops	Test photos	Weight in the loss
Projectile	4,465	672	984	221	0.49
Grenade	3,105	514	650	213	0.70
Mortar bomb	2,387	453	559	118	0.91
Cartridge	738	100	149	6	2.96
Aviation bomb	224	44	65	34	9.75
Total	10,919	1,783	2,407	522	

Note. From notebook Sections 2.3, 2.4 and 3.3. One crop per labelled item. The test-photo column counts the test photos that hold at least one crop of the category; its total counts each photo once. The weight in the loss is explained under Training.

The Task and How the Data Support It

The task is supervised multi-class image classification: given a 64 × 64 crop around one labelled item, predict which of the five categories it belongs to. Each crop is a square around the item's box, widened by 15% of the box's longer side on each side to keep some context, filled with grey where it runs past the edge of the photo (23.6% of crops), and shrunk to 64 pixels, a size set to keep a training epoch under a minute on a two-core processor. Real scenes labelled by an expert suit this task, because the model has to learn the categories from items as they are found, with vegetation, soil, corrosion and other items around them (Figure 1). They also come close to the setting where such a tool would be used, war-torn areas and former battlefields, since most of the photos show remnants of Europe's historical conflicts as they were found in the field (Craioveanu et al., 2026). Two properties of the data limit what it supports. Every photo holds at least one labelled item, and none of the 12 categories is battlefield debris, which is not UXO,

mines or other explosive hazards but is found with them on former battlefields, so a model trained on it can say which category an item looks most like, but not whether it is ordnance at all. And the items are small in their photos: half of the boxes cover less than 1.04% of the photo.

Data Quality

Two checks shaped the evaluation. First, near-duplicate photos: comparing a 64-bit difference hash of each photo, I found 123 pairs of photos within 4 bits of each other, 99 of them with identical hashes, which in the pairs checked by eye were the same photo. The creators removed near-duplicates in two stages before the split, so that photos of the same item or background would not be spread across the splits (Craioveanu et al., 2026), but 31 test photos still have a near-duplicate in the training or validation split, and they hold 57 of the 2,407 test crops. Such overlap makes a test easier than it looks (Barz & Denzler, 2020), so I scored the models again without them. In 6 of the 99 identical pairs, the two copies carry different numbers of labels. Second, crops cluster in a few photos. The median photo holds one crop, but the 10 largest test photos hold 39% of the test crops, and the 149 test cartridges come from only 6 photos, 3 of which hold 88% of them. Crops from one photo share its scene, light and camera, so they are not independent examples, and I treated the photo, not the crop, as the unit when estimating how sure the results are.

Model Architecture and Design Decisions

Why a CNN

A CNN suits images because its convolution layers learn small local patterns, such as edges and outlines, with the same weights at every position in the image, and build larger shapes from them layer by layer (Goodfellow et al., 2016). This is the brief’s image-based task type. The network is defined in full in PyTorch (Paszke et al., 2019) and trained from scratch, with no

weights learned elsewhere, so that the comparison tests the training change and nothing borrowed.

The Baseline Architecture

The baseline has four convolution blocks, each a 3 × 3 convolution, batch normalisation, a ReLU activation and 2 × 2 max pooling, followed by global average pooling, dropout and one linear layer with five outputs: 390,181 weights in all.

- **Depth and width.** The blocks double the channels (32, 64, 128 and 256) while pooling halves the image (64, 32, 16, 8 and 4 pixels), trading resolution for richer features. Stacked small 3 × 3 kernels see a wide area with few weights (Simonyan & Zisserman, 2015). Four blocks bring a 64-pixel crop down to a 4 × 4 map, about as deep as an input of this size allows.
- **Batch normalisation** rescales each channel's outputs during training, which steadies and speeds up learning (Ioffe & Szegedy, 2015); the convolutions therefore need no bias term.
- **ReLU** keeps positive values and sets negative ones to zero, the standard non-linearity in CNNs (Goodfellow et al., 2016).
- **Global average pooling** averages each of the 256 final maps to one number, so the classifier has few weights and does not depend on where in the crop a feature appears (Lin et al., 2014).
- **Dropout** sets 30% of those 256 numbers to zero at random during training, which discourages the network from relying on a few features (Srivastava et al., 2014).
- **Output.** Five scores, one per category, which a softmax turns into probabilities; the highest probability is the model's confidence.

The network is small on purpose: about 11,000 training crops would not support a large network trained from scratch, and the whole notebook, with two models, had to run on an ordinary computer's processor in about half an hour.

Training

- **Loss.** Cross-entropy, the standard loss for choosing one class among several, with each category weighted by the inverse of its share of the training crops so that every category counts equally in total (Table 1). Weighting the loss is a common way to keep a network from ignoring rare classes (Johnson & Khoshgoftaar, 2019), but it moves the model's decisions towards them, which raises their recall at a cost in precision.
- **Optimiser and schedule.** Adam (Kingma & Ba, 2015), starting from a learning rate of 0.001, lowered along half a cosine curve to almost zero by the 20th epoch (Loshchilov & Hutter, 2017). The schedule and the number of epochs were set from pilot runs judged on validation data only: at a constant rate, validation scores were still rising and jumping from epoch to epoch at epoch 15, and with the schedule they levelled off in the second half of training.
- **Light augmentation, in both models.** Each training crop is flipped and turned by a random multiple of 90 degrees, since an item can lie in any direction.
- **Model selection.** After each epoch, each model was scored on the validation crops, and the weights from the epoch with the best macro-F1 were kept; the test crops played no part in any choice.
- **Reproducibility.** Every random step is seeded, PyTorch uses deterministic algorithms, and training runs on the processor with two threads, so a rerun on the same computer

gives the same numbers. Each model trained in about 14.5 minutes on a two-core processor.

Experimental Comparison

The Change

The second configuration changes one thing, random erasing in training. For each training crop, with a probability of 0.5, it picks a rectangle covering 2% to 40% of the crop, with a height-to-width ratio between 0.3 and 3.3, and fills it with random values, the default settings of the original method (Zhong et al., 2020). Table 2 sets out what changed and what stayed the same. The two models start from the same weights and see the same crops, in the same order, with the same flips and turns, because each source of randomness has its own seeded generator and random erasing draws from a separate one.

Table 2

What Changed and What Stayed the Same

Element	Baseline	Second configuration
Architecture (390,181 weights)	Same	Same
Starting weights, batch order, flips and turns	Same seeds	Same seeds
Loss and class weights, optimiser, schedule, batch size, 20 epochs	Same	Same
Model selection on validation macro-F1	Same	Same
Training crops	Flipped and turned	Flipped and turned, with a random rectangle hidden in half of them

Note. From notebook Section 6.1.

Why It Is a Meaningful Comparison

In my experience as an EOD technician, the features that tell one model of ammunition from another are often found in a part of the item that is obscured in the photo, because it is covered in brush or partially buried, and the nuanced, slight differences between items can change how they function and so the hazards involved. Random erasing was designed to reduce

over-fitting and make a network robust to occlusion by training it on images with parts erased (Zhong et al., 2020), so it directly tests whether training with hidden parts helps a model when parts are hidden. It is a regularisation strategy, one of the kinds of change the brief allows, and it leaves the architecture and every other setting as they were.

How the Two Models Were Compared

I tested both models on the 2,407 clear test crops and on the same crops with one square patch hiding 10%, 25% or 40% of each, at random but fixed positions, identical for both models. The patch is a flat square of the training crops' average colour, a simple stand-in for brush, soil or shadow that differs from the noise-filled rectangles of training, so the second model was not shown its test conditions in advance. The main score is macro-F1, the mean of the five categories' F1 scores, which weights every category equally rather than letting the largest dominate (Sokolova & Lapalme, 2009); I also report accuracy, and precision, recall and F1 for each category. Because the crops cluster in photos, I estimated how sure each result is with a bootstrap that resamples whole test photos (Efron & Tibshirani, 1993; Field & Welsh, 2007): 2,000 times, as many photos as the test split holds were drawn at random with replacement, and both models were scored on the same draws, which gives 95% intervals for each score and for the difference between the models. Finally, since every answer goes to a technician, I set a confidence cut-off on the validation crops at which the answers kept were 95% right, and crops below it are reported as "unknown", a form of selective classification (Geifman & El-Yaniv, 2017).

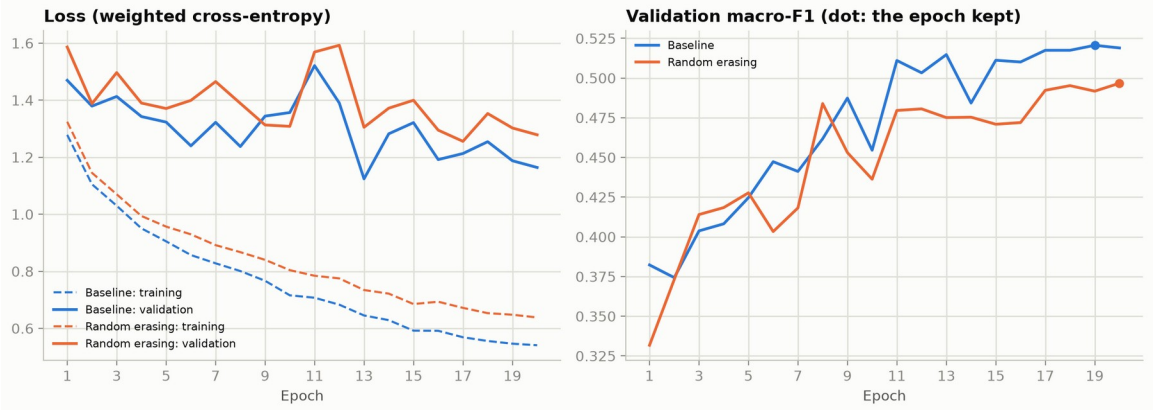
Results and Interpretation

Training Behaviour

Both models learned steadily (Figure 2). The baseline's training loss fell from 1.278 in the first epoch to 0.541 in the last, and its validation macro-F1 rose from 0.382 to 0.511 by epoch 11 and then held between 0.48 and 0.52 as the learning rate shrank; its best epoch, 19, scored 0.521. Its validation loss jumped from epoch to epoch and ended at 1.164, about twice the training loss: the network fitted the training crops much better than crops from new photos, a sign that some of what it learned belongs to the training scenes rather than to the items, which may include the shortcuts that Geirhos et al. (2020) describe, rules that work on the training data but fail to transfer. With random erasing, the training loss stayed higher throughout (0.638 in the last epoch), as expected when part of half the crops is hidden, and on the validation crops the baseline scored higher in 16 of the 20 epochs, including every epoch from the ninth; the random-erasing model's best was 0.497, at epoch 20.

Figure 2

Training and Validation Curves of the Two Models



Note. From notebook Section 6.3. Left: the weighted cross-entropy loss on the training crops (dashed) and the validation crops (solid). Right: validation macro-F1; the dot marks the epoch

whose weights were kept. The random-erasing model's training loss is computed on erased crops.

Test Scores on Clear Crops

On the test crops the order reversed: the random-erasing model reached an accuracy of .678 and a macro-F1 of .550, against .646 and .509 for the baseline (Table 3). Most of its gain came from projectiles and cartridges. Grenades were the easiest category for both models. The baseline called 24% of the test projectiles mortar bombs and the random-erasing model 14%, and both called 14% of the mortar bombs projectiles (Figure 3). That confusion matters less in the field than the numbers suggest: technically, a mortar is a subclassification of a projectile, and in my experience the two generally generate the same hazards. Cartridges and aviation bombs were the hardest categories. Most cartridges were missed, scattered across every other category, and they come from only 40 photos in the whole dataset. Aviation bombs show the cost of the class weighting: the models found nearly three in four of them, but only about one crop in seven that they called an aviation bomb was one, because the loss counted each aviation-bomb crop about 20 times as much as a projectile crop. Shifting a model's decisions towards a rare class, as class weights and adjusted decision thresholds both do (Buda et al., 2018; Johnson & Khoshgoftaar, 2019), raises that class's recall and lowers its precision, as here.

Table 3

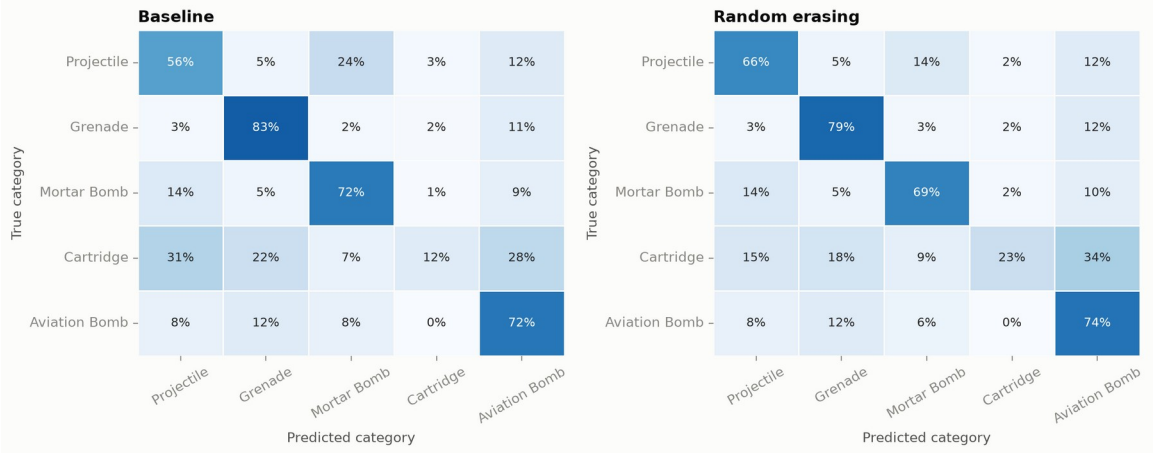
Precision, Recall and F1 on the Clear Test Crops

Category	Test crops	Baseline precision	Baseline recall	Baseline F1	Erasing precision	Erasing recall	Erasing F1
Projectile	984	.793	.563	.658	.834	.664	.739
Grenade	650	.826	.826	.826	.817	.788	.802
Mortar bomb	559	.602	.716	.654	.688	.687	.688
Cartridge	149	.316	.121	.175	.415	.228	.294
Aviation bomb	65	.140	.723	.234	.134	.738	.227
Accuracy	2,407			.646			.678
Macro-F1	2,407			.509			.550

Note. From notebook Section 7.1. “Erasing” is the model trained with random erasing. Precision is the share of crops given a category that belong to it; recall is the share of a category’s crops given that category.

Figure 3

Confusion Matrices on the Clear Test Crops



Note. From notebook Section 7.2. Each row shows how the test crops of one category were labelled, as shares of the row; the diagonal is the recall.

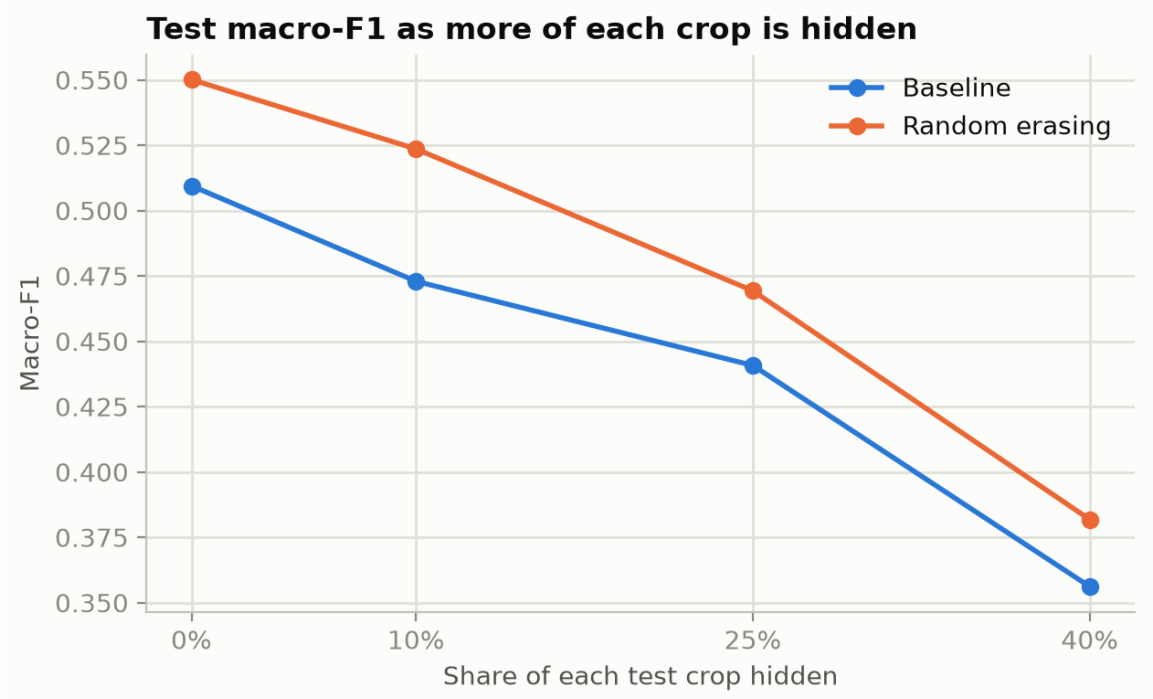
Partly Hidden Crops

Hiding part of each crop lowered both models’ scores (Figure 4): from clear crops to 40% hidden, macro-F1 fell by 0.153 for the baseline and by 0.168 for the random-erasing model. The

random-erasing model stayed ahead at every level, but its lead did not grow as more was hidden: 0.041 on clear crops, then 0.051, 0.029 and 0.026 with 10%, 25% and 40% hidden (Table 4). So in this experiment, training with hidden rectangles did not make the model hold up better when part of an item was hidden; whatever it gained, it gained about equally with and without the patch. This fits the finding that standard CNNs have only limited robustness to partial occlusion (Kortylewski et al., 2021). One possible reason, which I did not test, is that the training rectangles are filled with random noise and the test patches with one flat colour, so the model may have learned to ignore noise rather than to cope with a missing part. Zhong et al. (2020) found random and mean-colour fills about equally effective on standard benchmarks, but I found no study comparing them under real occlusion.

Figure 4

Test Macro-F1 as More of Each Crop Is Hidden



Note. From notebook Section 7.3. The same 2,407 test crops with one square patch hiding 10%, 25% or 40% of each, at the same positions for both models.

Table 4

Test Macro-F1 With 95% Intervals From Resampling the Test Photos

Test crops	Baseline	Random erasing	Difference	Resamples where erasing scores higher
Clear	.509 [.446, .586]	.550 [.475, .616]	+.041 [−.018, +.084]	89%
10% hidden	.473 [.416, .532]	.524 [.448, .603]	+.051 [+.002, +.104]	98%
25% hidden	.441 [.381, .485]	.469 [.396, .545]	+.029 [−.022, +.107]	82%
40% hidden	.356 [.284, .403]	.382 [.315, .422]	+.026 [−.020, +.085]	82%

Note. From notebook Section 7.4. 2,000 resamples of the 522 test photos that hold crops of the five categories, each keeping all the crops of the photos drawn, with both models scored on the same resamples. The intervals are the 2.5th and 97.5th percentiles; the difference is random erasing minus the baseline.

How Sure the Difference Is

The intervals from resampling the test photos show how fragile these differences are (Table 4). Each model's macro-F1 spans a band about 0.10 to 0.15 wide, because a handful of photos carry much of the score, and the 95% interval for the difference includes zero on clear crops and with 25% and 40% hidden, and only just excludes it with 10% hidden. Those intervals cover only the choice of test photos. Training adds variation of its own: an earlier full run of the same notebook, identical except that the crops were stored in a different memory order, which changes the order of some floating-point sums, ended with a difference of +0.020 on clear crops instead of +0.041, and −0.011 with 40% hidden instead of +0.026. Between two runs of the same experiment, the gap between the models moved by up to 0.045, as much as the gap itself. Such instability is known: even a one-bit change in a network's starting weights can lead training to a very different model, which makes small improvements hard to tell from run-to-run variability (Summers & Dinneen, 2021), and PyTorch does not guarantee identical results across platforms (PyTorch Contributors, 2026). Removing the 31 test photos with near-duplicates changed either

model's macro-F1 by less than 0.005, so the near-duplicates do not explain the scores. Taken together, the experiment gives no reliable evidence that random erasing helps this task: the test scores favour it slightly, the validation scores favour the baseline, and the gap is within the noise of a single training run of each model.

Confident Answers and "Unknown"

To keep the answers it gave 95% right on the validation crops, each model needed a confidence of about 0.96, and it kept only 19% (baseline) and 16% (random erasing) of the validation crops. On the clear test crops, the same cut-offs sent 84% and 85% of the crops to "unknown", and the models were right on 97% and 98% of the rest. With a quarter of each crop hidden, almost all crops (98% and 97%) went to "unknown", and the few kept were still right 96% and 98.5% of the time, so the models' confidence fell when part of the item was hidden, which is the behaviour wanted. Of the 340 crops from the seven categories the models never saw, only 2% passed the cut-off. The softmax confidence of a neural network is not a calibrated probability (Guo et al., 2017), but it is usually lower for wrong and unfamiliar inputs than for right ones (Hendrycks & Gimpel, 2017), which is what the cut-off relies on; the few unfamiliar items that passed it, among them 2 of the 29 landmines for the baseline, show that it is a filter, not a guarantee.

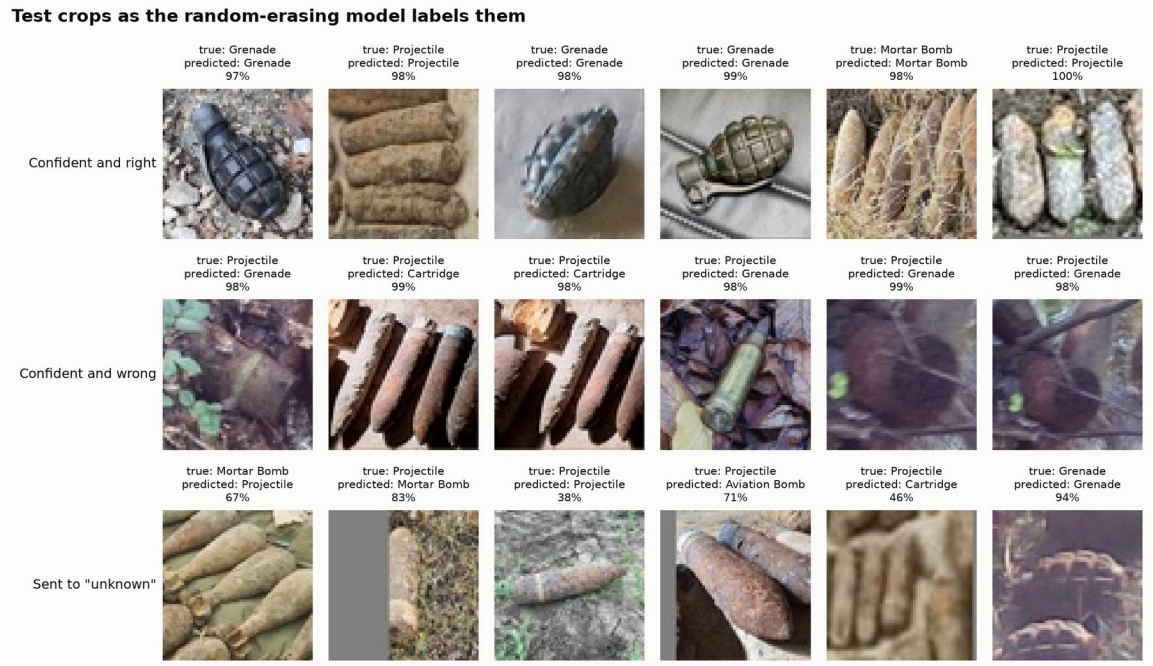
An Example of the Model's Behaviour

The most telling errors are the confident ones (Figure 5). The random-erasing model passed on 358 test crops with confidence; 351 were right and 7 were wrong. All six wrong ones shown are labelled projectiles, and I checked that the dataset's labels on them are correct. Two neighbouring items in one row were called cartridges at 98% and 99%, a green item among leaves was called a grenade at 98%, and three dark crops that appear to come from one scene

were called grenades at 98% and 99%. Many of the confident mistakes are just obscured or partial views of items. In those three, maybe 25% of the projectile is being seen; the rest of it is obscured by vegetation, and what shows is the base of the projectile. That is exactly the condition this experiment was about, real occlusion by vegetation, and the model trained with random erasing still met it with confident wrong answers. The crops it sent to “unknown” show the other side: because I have dealt with a lot of UXOs that have been sitting in the ground for long periods, I can see that these are clearly UXOs that have sat and aged in the ground, of the kind seen in Belgium and other places, and I could name each of them.

Figure 5

Test Crops as the Random-Erasing Model Labels Them



Note. From notebook Section 7.7. Six crops drawn at random from each group; each title gives the dataset’s label, the model’s answer and its confidence. The cut-off for a confident answer is 0.966. Photos from CTX-UXO (Craioveanu & Stamatescu, 2025), CC BY 4.0.

In Plain Terms

For a reader without a technical background: the model looks at a small picture of a single item of unexploded ordnance and suggests which of five kinds it is. It is right about two times in three overall, and much more often when it is sure of itself: when it is confident enough to answer, it is right about 97 times in 100, but it is that confident for only about one item in seven, and for the rest it says “unknown”. Training it with parts of its pictures blanked out, to prepare it for items half hidden by plants or soil, did not reliably help, and its most serious mistakes were confident answers about items mostly hidden by vegetation, the case where a wrong answer would matter most. It has also never been shown battlefield debris, so it cannot tell whether an item is ordnance at all. I would say that this is a model to enhance the speed of workflow and one of the first stages in enhancing scanning techniques through drones to do quick surveillance of fields. It never states that an item is safe, and every answer goes to a technician.

Limitations and Risks

- **One country’s photos, taken by experts.** All the photos come from Romania, of items from Europe’s historical conflicts, taken by trained personnel, some during training sessions and some with items gathered in rows and piles. On a larger dataset I would expect to see lots of different environments and lots of different lighting conditions, particularly lots of items in sand and lots of partial components of ammunition. Networks often learn the backgrounds and settings of the places they were trained on and fail elsewhere (Beery et al., 2018; Geirhos et al., 2020), so these results may not carry over to photos from drones, survey teams or local people in other regions.

- **No battlefield debris.** Every crop shows a labelled item of ordnance. On the former battlefields where this tool would be used, battlefield debris, which is not UXO, mines or other explosive hazards, is found alongside the hazards, and these models were never trained on any: they would give each piece of debris one of the five categories, or "unknown". They cannot answer the first question in the field, whether an item is an explosive hazard at all, and their answers only make sense for an item already suspected to be one.
- **Overlapping categories.** A mortar is technically a subclassification of a projectile, but the labels treat the two as separate classes, so the scores count as errors some answers that matter little in practice.
- **Few photos behind some categories.** The 149 test cartridges come from 6 photos and the 65 test aviation bombs from 34, so their scores rest on a few scenes, and the intervals in Table 4 are wide.
- **Near-duplicates and uneven labels.** 31 test photos have near-duplicates in training or validation despite the creators' deduplication, and in 6 pairs of identical photos one copy has more labels than the other. Their effect on the scores was under 0.005 here, but similar overlaps have changed results on widely used benchmarks (Barz & Denzler, 2020).
- **A small model at low resolution.** At 64 × 64 pixels many crops are enlarged from small boxes and lose the fine details that tell items apart, and a network trained from scratch on about 11,000 crops learns less than one that starts from features learned on millions of images.

- **One run of each model.** The difference between the models was as large as the difference between two runs of the same experiment, so one run of each configuration cannot settle the comparison (Summers & Dinneen, 2021), and on another computer the scores can differ somewhat (PyTorch Contributors, 2026).
- **A stand-in for occlusion.** The test patches are flat squares of one colour, not brush, soil or corrosion, so the hidden-patch tests measure only one simple form of hiding; the confident mistakes in Figure 5 show the real form.
- **Uncalibrated confidence.** The cut-off was set on validation crops that are themselves clustered in a few photos, and softmax confidence is not a calibrated probability (Guo et al., 2017), so the 95% target may not hold on new photos.

Ethical and Responsible Use

Over-Reliance on Confident Answers

If technicians became too reliant on the model's confidence in an answer, this could lead to misclassification, which would then lead to mishandling and can then cause injuries. The risk is well documented. Automation bias, over-reliance on a decision aid, produces errors of both omission and commission, occurs in both novices and experts, and cannot be prevented by training or instructions (Parasuraman & Manzey, 2010). In a study of mammography reading, radiologists' correct ratings fell from about 80% to between 20% and 46% when the suggestions of an AI-based system they were shown were wrong, including among very experienced readers (Dratsch et al., 2023). The confident mistakes in Figure 5 show how the same could happen here: a confident answer about a mostly hidden item is exactly where a technician might be tempted to trust the model. I am not sure yet of the best way to prevent this. A review of automation bias points to measures such as emphasising the user's accountability, presenting information rather

than recommendations, showing confidence levels with each answer, and training (Goddard et al., 2012). The design already takes some of these steps: the model never says that an item is safe, it shows its confidence, it says “unknown” when unsure, and every answer goes to a technician as a first indication to confirm.

Asking the Public to Take Photos

Photos could also come from local people who find a suspected item. It is not that people would be encouraged to go and seek these items, but if they do, a tool like this would give them the ability to take pictures and have the item confidently cleared by an EOD technician, on the same thread as the applications that let people self-check for skin cancer. A warning should go out with the application, though, not to handle the item; it is meant only for a picture. The analogy also shows the risk. A systematic review found that algorithm-based smartphone apps for skin cancer could not be relied on to detect all cases, that they are likely to perform worse when used by their intended users, and that the process for awarding them CE marking did not adequately protect the public (Freeman et al., 2020). A public UXO app that gave a confident or reassuring answer could be read as “safe” and lead someone to handle the item. Any such tool would therefore need to fit within explosive ordnance risk education, in which safety messages are at the heart of every initiative and people are told whom to report to when they find explosive ordnance (United Nations Mine Action Service [UNMAS], 2020): it should send every report to the responsible authority, never say that an item is safe, and carry the warning not to touch.

Who Is Served Well, and Controlled Information

A model trained on these photos would serve some places better than others. It has learned from one country’s soils, vegetation and seasons, and from Europe’s historical munitions, so people in regions unlike Romania, such as those where items lie in sand, would get

worse answers, often without anyone knowing. A second risk lies in the data a stronger model would need: some of the most useful sources, such as national EOD technical publications, are restricted, and a model or dataset built from them could spread controlled information beyond the people entitled to it. Such material should stay within authorised channels, and published work, like this report, should stay at the level of categories and counts.

Future Improvements

- **More and wider data.** Data that would be useful could be field reports taken by technicians, photos from the collections of different museums, and UXO databases such as CAT-UXO, or material from the Geneva International Centre for Humanitarian Demining (GICHD) or UNMAS; websites such as Bulletpicker could also be drawn on, and governments and government organisations could be approached for material such as the United States' EOD technical publications, the 60-series. All of these come with complications in getting them and then cleaning them first. CAT-UXO's terms allow its pages to be copied for personal use (CAT-UXO, n.d.), GICHD's allow personal and informational use without modification (Geneva International Centre for Humanitarian Demining [GICHD], 2019), the United Nations allows non-commercial use of the International Mine Action Standards with attribution (UNMAS, 2020), and access to the joint EOD technical manuals is limited to EOD-qualified personnel performing EOD duties (Department of the Army, 2005), so each would need its owner's permission, and restricted material could only ever be used within authorised channels, never in a public dataset. The creators of CTX-UXO also note that public repositories often show mock-ups or items in perfect condition on white backgrounds, which make models less robust in real-world scenarios (Craioveanu et al., 2026), and museum photos raise the same

issue. New data should add what the current set lacks: many environments and lighting conditions, items in sand, partial components, and views from drones.

- **A model for battlefield debris.** Additional training models would be useful to scope battlefield debris, which is not UXO, mines or other explosive hazards. The dataset's creators already recommend adding at least 500 photos of empty backgrounds, such as terrain, soil and snow, as negative examples (Craioveanu et al., 2026); photos of debris from former battlefields would go further and let a model give a first indication of whether an item is ordnance at all, the question the current models cannot answer. An answer of "debris" would carry the same risk as any reassuring answer, so it too would go to a technician to confirm, never as a statement that an item is safe.
- **A specialist in the loop.** As technicians confirm or reject the model's answers, their decisions can become new labels that refine it, and they build the record of field performance needed before relying on it.
- **A stronger model.** A network pretrained on ImageNet usually transfers well to new image tasks, although the benefit can be small for fine-grained categories (Kornblith et al., 2019), so it should be tested here. Larger crops would keep more detail, and detecting items in whole photos, as the dataset's creators do (Craioveanu et al., 2026), would remove the need for boxes drawn by people. Labels that nest categories, such as mortar bombs within projectiles, would let a model answer "projectile" when it cannot tell which kind.
- **A firmer experiment.** Several training runs of each model with different seeds would separate the effect of random erasing from run-to-run variation (Summers & Dinneen, 2021). Testing on items photographed partly hidden by vegetation or soil, and comparing

training rectangles filled with noise and with a flat colour, would test occlusion directly, and architectures built for partial occlusion are another route (Kortylewski et al., 2021). Calibrating the confidence before setting the cut-off, for example by temperature scaling (Guo et al., 2017), would make the “unknown” band more dependable.

- **Towards drones.** This model is one of the first stages in enhancing scanning techniques through drones to do quick surveillance of fields. Deep learning has already found surface-laid mines in drone photos of test sites (Baur et al., 2020), and a classifier like this one could give a first indication for each item a drone finds.
- **A trial before use.** Before any use in the field, the model would have to show a reasonable confidence level and then be tried for a period, demonstrating that teams were actually more efficient with it.

Use of AI Tools

I used Claude (Anthropic). I ran the notebook and checked its results.

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